Project Design Phase
Proposed Solution

Date	15-07-2026
Team ID	SWTID-2026-1581
Project Name	Credit Card Approval Prediction System
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	**Problem Statement (Problem to be solved)**	Banks and financial institutions currently assess credit card applicants through slow, manual review that relies on scattered records (income, employment, past credit behaviour). This leads to long waiting times for applicants and inconsistent, reviewer-dependent risk decisions for the bank, increasing the chance of approving risky applicants or rejecting creditworthy ones.
2.	**Idea / Solution description**	A web-based Credit Card Approval Prediction System where an applicant fills in personal, financial, family and credit-history details in a simple form. The system preprocesses the inputs and runs them through a trained machine-learning model (best of Logistic Regression, Decision Tree, Random Forest and XGBoost, selected by F1-score) to instantly classify the applicant as low risk (Approved) or high risk (Rejected).
3.	**Novelty / Uniqueness**	Combines demographic, financial and historical credit-repayment data into a single, consistent, data-driven risk score instead of ad-hoc manual judgment. Compares multiple ML algorithms and automatically selects the best-performing one, and the entire pipeline (preprocessing, model, encoders) is packaged for instant, repeatable predictions rather than one-off manual analysis.
4.	**Social Impact / Customer Satisfaction**	Applicants get an instant decision instead of waiting days, reducing anxiety and uncertainty. Consistent, data-driven scoring reduces reviewer bias and treats similar applicant profiles similarly. Banks can serve more applicants with the same staff, and can extend responsible credit access to more creditworthy customers who might otherwise be overlooked by rigid manual rules.
5.	**Business Model (Revenue Model)**	Offered as a B2B SaaS risk-scoring module that banks and NBFCs license/subscribe to (per-prediction or monthly subscription pricing), integrated into their existing card-issuance workflow via API. A freemium tier could let fintech startups test the model on limited volume before scaling to a paid plan.
6.	**Scalability of the Solution**	The stateless Flask design and lightweight pickle-based inference allow the app to be scaled horizontally on Render (or any cloud platform) by adding more instances as applicant volume grows. The model can be retrained on new data and redeployed independently of the web layer, and the same architecture can be extended to other credit products (loans, BNPL) with minimal changes.